CHAPTER 8

DATABASE MANAGEMENT SYSTEM

CHAPTER ESSENCE AT A GLANCE:
Database Management Systems overcome file processing limitations by providing structured three-level data abstraction, relational schema integrity, centralized access control, and mathematical relational algebra operations.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Limitations of File Systems vs DBMS Advantages:** Traditional file systems suffer from data redundancy, inconsistency, difficulty in data access, concurrency anomalies, and security silos, which DBMS eliminates through centralized relational management.

• **Three-Level ANSI-SPARC Data Abstraction:** The database architecture isolates internal physical storage structures, conceptual logical enterprise schemas, and external user views to ensure physical and logical data independence.

• **Relational Data Model Architecture:** Data is structured into relations characterized by fixed attribute columns (degree) and variable tuple rows (cardinality), governed by atomic domains and absence of duplicate tuples.

• **Relational Keys and Integrity Constraints:** Candidate, primary, and alternate keys uniquely identify relation entities, while foreign keys establish referential integrity between child and parent relations.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{Relational Metrics} = \text{Degree} = \text{Total Attributes (Columns)}, \quad \text{Cardinality} = \text{Total Tuples (Rows)}$$

Fundamental Relational Schema Dimensions

$$\text{Data Independence Tiers} = \frac{\text{Logical Independence: External Views Shielded from Conceptual Schema Changes}}{\text{Physical Independence: Conceptual Schema Shielded from Physical Storage Changes}}$$

ANSI-SPARC Data Independence Model

$$\text{Fundamental Relational Algebra} = \sigma_{\text{Condition}}(R)\ \text{(Selection)}, \quad \pi_{\text{Attributes}}(R)\ \text{(Projection)}, \quad R \times S\ \text{(Cartesian Product)}$$

Relational Algebraic Operations

$$\text{Integrity Invariants} = \frac{\text{Entity Integrity: Primary Key} \neq \textbf{NULL} \text{ and Strictly Unique}}{\text{Referential Integrity: Foreign Key} \in \text{Parent Primary Key Values} \cup \{\textbf{NULL}\}}$$

Core Relational Integrity Constraints

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Never confuse Degree with Cardinality: Degree is the number of attributes (columns) in a table; Cardinality is the total number of tuples (rows).

• **Alert:** Entity integrity strictly forbids primary key attributes from containing NULL values; a primary key must be unique and non-null for every tuple in the relation.

• **Alert:** A foreign key value must either match an existing primary key value in the referenced parent relation or be set to NULL; orphaned foreign key references violate referential integrity.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State four major disadvantages of the traditional file processing system that are overcome by a DBMS. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Data Redundancy and Inconsistency: Duplicate copies of the same data across files leading to conflicting information when updated. **[1 Mark]**

• Data Isolation and Integrity Problems: Data scattered in various file formats making coordinated retrieval difficult and lacking automatic integrity constraint checks. **[1 Mark]**

2. Define the terms Degree and Cardinality of a relation. If a table has 5 columns and 20 rows, find its degree and cardinality. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Degree: The total number of attributes (columns) present in a relation. In this table, Degree = 5. **[1 Mark]**

• Cardinality: The total number of tuples (rows or records) stored in a relation. In this table, Cardinality = 20. **[1 Mark]**

3. Explain the three levels of data abstraction in a database system with a labeled block diagram. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Physical Level (Internal): The lowest level that describes how data is physically stored on storage media (record layouts, disk indices, byte offsets). **[1 Mark]**

• Logical Level (Conceptual): Describes what data is stored in the entire database and the structural relationships among entities (table schemas, data types, constraints). **[1 Mark]**

• View Level (External): The highest level that describes customized sub-schemas and user-specific views of the database, hiding confidential fields. **[1 Mark]**

4. Distinguish between Candidate Key, Primary Key, and Foreign Key in a relational database system with an example. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Candidate Key: Any minimal set of attributes that uniquely identifies every tuple in a relation (e.g. AdmissionNo and EmailId in Student). **[1 Mark]**

• Primary Key: The specific candidate key chosen by database designers to uniquely identify records; strictly unique and never NULL (e.g. AdmissionNo). **[1 Mark]**

• Foreign Key: An attribute in a child table whose values reference the primary key of a parent table, establishing referential integrity (e.g. CourseId in Student referencing Course table). **[1 Mark]**

5. Differentiate between Physical Data Independence and Logical Data Independence. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Physical Data Independence: The capacity to modify physical storage schemas (such as adding indexes or altering disk file structures) without having to rewrite conceptual schemas or application programs. **[1.5 Marks]**

• Logical Data Independence: The capacity to modify conceptual schemas (such as adding new attributes or tables) without altering existing external view levels or requiring application rewrites. **[1.5 Marks]**

6. Explain fundamental relational algebra operations: Selection (sigma) and Projection (pi) with syntax and tabular examples. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Selection (sigma): A horizontal unary operation that selects a subset of tuples from a relation satisfying a specified predicate condition (e.g. sigma(Marks > 80)(Student)). **[1.5 Marks]**

• Projection (pi): A vertical unary operation that extracts specified attribute columns from a relation, discarding all other columns and removing duplicate rows (e.g. pi(Name, Course)(Student)). **[1.5 Marks]**

7. Consider a database with tables Student(AdmNo, Name, CourseId) and Course(CourseId, CourseName, Fees). Explain how Primary Keys and Foreign Keys enforce Entity and Referential Integrity. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Primary Key Specification: CourseId in Course table and AdmNo in Student table serve as Primary Keys, uniquely identifying each course and student respectively. **[1 Mark]**

• Entity Integrity Rule: Neither AdmNo in Student nor CourseId in Course can be NULL or duplicate, ensuring every entity record is distinguishable. **[1 Mark]**

• Foreign Key Binding: CourseId in Student table is designated as a Foreign Key referencing Course(CourseId) in the parent Course table. **[1 Mark]**

• Referential Integrity Rule: A student cannot be enrolled with a CourseId that does not exist in the Course table; deleting a course referenced by students is restricted or cascaded. **[1 Mark]**